Personal Care SUPPLY CHAIN ANALYSIS REPORT

1. Executive Summary

This report analyzes a 100-SKU supply chain dataset spanning three personal care categories — haircare, skincare, and cosmetics — across five Indian cities. The dataset captures a single snapshot per product (no date field is present), covering the full chain from manufacturing and quality inspection through to warehousing, shipping, and final sale.

Across the 100 SKUs, the business generated $577,604.8 in revenue from 46,099 units sold, against 4,777 units currently held in stock. Average cost to ship a unit is $5.55, and the average supplier lead time is 17.1 days.

Skincare is the strongest category by revenue and volume despite not having the most SKUs, while cosmetics is the smallest contributor on every measure. The clearest operational risk in the data is quality: only 23% of SKUs passed inspection, with 41% still pending and 36% failed.

2. Key Performance Indicators

KPI	Value
Total Revenue Generated	$577,604.8
Total Products Sold	46,099 units
Total Stocks	4,777 units
Average Shipping Costs	$5.55 / unit
Average Lead Time (procurement)	17.1 days

Note: *the dataset contains two similarly-named lead time fields. "Average Lead Time" here refers to the original "Lead time" column (interpreted as supplier/procurement lead time). A second field, "Lead times" (order fulfillment lead time, average 15.96 days), was excluded from the KPI cards to avoid duplicating the same concept on the dashboard.*

3. Findings by Business Area

3.1 Revenue & Volume by Product Type

Product Type	SKUs	Revenue Generated	Production Volume
Skincare	40	$242,000	24,366
Haircare	34	$174,000	19,957
Cosmetics	26	$162,000	12,461

Skincare leads on every measure — revenue, production volume, and SKU count — making it the core category of the portfolio. Cosmetics has the fewest SKUs and the lowest output on every metric, suggesting it is either a newer line or a lower investment priority relative to the other two categories.

3.2 Performance by Location

Location	SKUs	Products Sold	Revenue Generated
Kolkata	25	12,770	$137,077.6
Mumbai	22	9,426	$137,755.0
Chennai	20	8,768	$119,142.8
Delhi	15	9,715	$81,027.7
Bangalore	18	5,420	$102,601.7

Mumbai generates almost the same revenue as Kolkata (~$137.8K vs ~$137.1K) from fewer units sold (9,426 vs 12,770) — a higher revenue-per-unit, worth investigating whether this comes from pricing, product mix, or fewer discounts in that location. Delhi sells a comparable volume to Mumbai but returns the lowest revenue of all five cities, making it the weakest location by revenue efficiency.

3.3 Quality: Inspection Results

Inspection Result	SKU Count	Share of Total
Pending	41	41%
Fail	36	36%
Pass	23	23%

This is the most important finding in the dataset. Fewer than a quarter of SKUs have a confirmed pass, more than a third have failed inspection outright, and 41% remain undecided. Any KPI on revenue or volume should be read alongside this — a meaningful share of the products driving the revenue totals above have not been confirmed to meet quality standards. Breaking this down by category, skincare and haircare each have a similar fail count (13 SKUs), while cosmetics has the lowest average defect rate (1.92%) versus haircare (2.48%) and skincare (2.34%).

3.4 Logistics: Transportation, Routes & Shipping

Transportation Mode	SKU Count	Share
Road	29	29%
Rail	28	28%
Air	26	26%
Sea	17	17%

Road, rail, and air are used at nearly equal rates, with sea freight the least common — consistent with a domestic-focused Indian distribution network rather than an import/export-heavy one. By route, Route A carries the most SKUs (43), followed by Route B (37) and Route C (20). Average shipping time is fastest for skincare (5.32 days) and slowest for cosmetics (6.58 days), meaning cosmetics both underperforms on revenue and takes the longest to ship — worth flagging as a compounding weakness for that category.

4. Data Notes & Limitations

- Snapshot, not a time series: the dataset has no date field, so nothing here reflects trend, seasonality, or month-over-month change — every figure is a static cross-section of 100 SKUs.
- Small sample size: 100 SKUs is enough to describe this dataset but not large enough to generalize confidently to a full catalog or a different time period.
- Two lead time fields: "Lead time" (procurement, 17.1 days avg.) and "Lead times" (fulfillment, 15.96 days avg.) are similarly named but not correlated in the data; this report treats them as two distinct stages of the supply chain, not duplicate values.
- "Costs" field: the dataset includes a generic "Costs" column ($52,924.6 total) whose exact scope (logistics-only vs. all-in cost) is not documented in the source; it is treated here as a general logistics cost estimate rather than a fully defined P&L line.

5. Recommendations

- Prioritize clearing the 41 pending inspections before drawing further conclusions from revenue and volume KPIs, since a large share of reported performance sits on unconfirmed quality.
- Investigate the cosmetics category specifically: lowest revenue, lowest production volume, and the slowest shipping times together suggest either a deprioritized line or an operational bottleneck worth a root-cause review.
- Review why Delhi converts sales volume into revenue less efficiently than Mumbai and Kolkata — this looks like a pricing or discounting difference rather than a demand problem, since unit volumes are comparable across the three cities.
- Treat this analysis as directional rather than final, given the sample size and single-snapshot nature of the data; revisit once transaction-level or time-stamped data becomes available.